

AI SEMICONDUCTOR INVESTMENT ANALYSIS

A Four-Layer Python Framework Combining Fundamental Screening, Portfolio Optimization, Technical Analysis and Forward-Looking Projections

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Executive Summary

This report presents a 4-layered analytical framework for evaluating AI semiconductor stocks, combining historical fundamental analysis, portfolio optimization, technical price action and forward looking projections. Data is sourced and analyzed through yahoo finance via yfinance library and python, and all models can be rerun at any time to reflect current market conditions. A top-down, bottom-up approach has been followed throughout the report.

PART 1 & 2 - Stock Screener & Portfolio Optimization

This section first breaks the AI industry into top 5 Sub-Sectors identified through external research. Using an ETF to represent each sub sector, it then measures the performance of each Sub-Sector on the basis of their yearly and 3 years annual returns. Once the best Sub-sector is established, the section dives into analyzing each company under the ETF. Primary assessment includes evaluation based off 8 core metrics. Which then leads to labelling the company's stocks as undervalued, fairly valued or overvalued, based on 3 other key metrics. Further, the stocks are assessed using metrics like Beta, Volatility, Sharpe Ratio, Correlation and Monte Carlo Simulation across 10,000 portfolio combinations to determine the optimal asset allocation across the 5 selected stocks. This part includes two variations of the optimal allocation, one including all 5 stocks picked earlier and one with just 4 of them.

PART 3 – Technical Analysis

Concepts like Price Volume Action, momentum and Wykoff Stage analysis are performed to identify the trends of each stock and timing of investment.

PART 4 – Forward Looking Analysis

Analyst consensus price targets and forward earnings projections provide a 12-month outlook for each stock, adding a forward-looking dimension to the report.

It is crucial to note that this report is only intended for education purposes and does not guarantee the same results. All the output is based on historical data. Past performances do not guarantee future profits.

PART 1: Stock Screener

STEP 1: Identification of top 5 AI Sub-Sectors and their analysis based off ETF Performances.

The AI industry is broken down into 5 major Sub-sectors. Each Sub-sector is represented by a publicly traded ETF, which represent a basket of companies within each sub-sectors, providing greater reliability than the usage of a single stock from each sector.

S.no	Sub-Sector	ETF	1 Year Return (%)	3 Year Returns (%)	Rank
1	Semiconductors & AI Chips	SOXX	149.71%	242.18%	1st
2	Broad AI	AIQ	50.29%	130.70%	2nd
3	Healthcare AI	ARKG	38.28%	-1.79%	3rd
4	Robotics & Automation	BOTZ	20.12%	31.91%	4th
5	Cloud & Data Centers	SKYY	16.14%	80.15%	5th

Interpretation & Analysis

Note: Values generated by Python are sourced using yfinance library, however, there may be slight differences in them compared to the values displayed on yahoo finance. While Yahoo uses the price return, our analysis has used the total return which accounts for dividends and stock splits giving a more accurate representation of figures.

Based on the ETF performance data, Semiconductors & AI Chips (SOXX) are identified as the strongest performing AI sub-sector across both timeframes in comparison to its peers, strongly carried and driven by demand of AI chips led by NVIDIA. Cloud and Data Centres show a solid 3-year performance, while holding a lower yearly return, indicating that the AI infrastructure industry has already priced in. Healthcare AI presents the most alarming signal, with a negative 3-year return followed by a sharp recovery of 38.28% on yearly returns suggesting the market bounce rather than continued growth.

Conclusion: SOXX is the best performing sub-sector. However, it is important to evaluate if the gains from massive run have already been priced in, or if it is still undervalued and therefore worthy of our investment, which is exactly what we will analyze in the coming step.

STEP 2: Fundamental Screening of 34 companies under SOXX

Following the identification of SOXX, all 34 companies included were evaluated using 8 key metrics. The table below depicts rationale behind the chosen metrics and the threshold used to finalize companies.

Metrics & Rationale

Metric	Rationale	Category	Thresholds
Trailing P/E	Universal Valuation tool. Lower P/E indicates a relatively attractive prices in relation to its current price.	Valuation	< 150
Forward P/E	Estimates future earnings relative to current prices. Lower forward P/E indicates that we might be buying higher future expected gains at a discount now, however this must be confirmed with other positive metrics like revenue & profit growth and ROE.	Valuation	< 45
EV/EBITDA	Unlike P/E, EV Multiple evaluates the companies value while being unaffected by its debt levels, tax rate or depreciation policies, therefore, providing value for a companies core operational performance. Lower EV multiple indicates a more attractive business.	Valuation	< 60

PEG Ratio	Pivotal for high growth companies as it assists in understanding if a high P/E ratio is truly justified and worth investing in.	Valuation/Growth	< 3
ROE	Calculates how efficiently the company is using shareholder's funds. It is a good measure of profit management. A higher ROE tells us that a company is able to generate more with less inputs, therefore being more attractive.	Quality	> 15%
Revenue Growth	In the AI space, markets are evolving and companies are continuously catching growth as they move. This forward-looking metric tells us how much the company is growing along in terms of catching trends and generating revenue from them.	Growth	> 10%
Profit Margin	Indicates a company's pricing power and operational efficiency. The higher the ratio, better it is for the firm.	Quality	> 10%
Debt-to-Equity	It is important to assess debt levels, especially for cyclical sectors like the semi-conductor industry. In bust periods, it might put a company into financial distress.	Risk	< 110

Together these 8 metrics present a complete 360-degree analysis of valuation, growth, quality and risk at the same time, giving us a better understanding to analyse and further shortlist stocks within the ETF.

Once the metrics for each company is established, a 2-tier screening process was carried through, with thresholds set for each metric. All thresholds were set based on live sector benchmarks derived from yahoo finance across all 34 companies in the SOXX, which can be rerun at anytime using the python file to reflect current conditions. To avoid distortion from extreme outliers all thresholds were calibrated against sector medians rather than means. In order to ensure the screening removes only truly problematic companies, thresholds were deliberately set generously above and below the given median levels.

The Final 5

9 companies out of 34 cleared all the benchmark rules, however, only 5 were chosen, based on AI relevance and diversification across the semiconductor sector.

Ticker	Company	P/E	F. P/E	ROE (%)	Rev. Growth	Profit Margin
MU	Micron Technology	42.01	8.00	39.82%	196.3%	41.49%
AVGO	Broadcom Inc.	61.71	19.23	37.28%	47.9%	38.85%
NVDA	NVIDIA Corporation	30.69	15.75	114.29%	85.2%	62.97%
LRCX	Lam Research Corp.	60.95	40.39	66.76%	23.8%	30.94%
SNDK	Sandisk Corporation	56.18	8.98	39.30%	251.0%	34.19%

STEP 3: Valuation Assessment

Three independent Valuation Metrics were applied to each of the 5 selected companies to ensure the prices are justified. Output is displayed through terms of undervalued, fairly valued and overvalued, with undervalued companies being the most attractively priced. The three metrics are as follows,

Methods	Metric	Rationale	Output Benchmarks
1	P/E Ratio vs the sector average	The P/E ratio is now compared to the Semi-Conductor sectors P/E ratio, which is 67.04 (derived from live python data sourced from yfinance). It shows how expensive a stock is compared to its peers right now.	15% below 67.04 -Undervalued 15% above 67.04 - Overvalued.
2	PEG Ratio	Assesses if a valuation of a company is justified by the its growth model	Below 1.0 - Undervalued 1.0 to 2.0 - Fairly Valued Above 2.0 - Overvalued ⁱ
3	Forward vs Trailing P/E	Displays the market's expectations about a company's stock in the future. Fulfills limitation of the other two methods relying on past data only. Companies will then be ranked based on the largest gaps between fwd and trailing P/E's.	Higher gap is considered better, as it depicts higher expectations of market for a significant growth ahead.

Ticker	Company	P/E vs Sector (Median: 67.04)	PEG Ratio	Fwd vs Trail P/E Gap	Final Verdict
MU	Micron Technology	Undervalued	Undervalued	81.5%	UNDERVALUED
AVGO	Broadcom Inc.	Fairly Valued	Undervalued	68.8%	UNDERVALUED
NVDA	NVIDIA Corporation	Undervalued	Undervalued	48.7%	UNDERVALUED
SNDK	Sandisk Corporation	Fairly Valued	No Data	84.0%	UNDERVALUED
LRCX	Lam Research Corp.	Fairly Valued	Fairly Valued	33.8%	FAIRLY VALUED

Interpretation & Analysis

Four out of five companies assessed as undervalued based on majority signals across the three methods of valuation used. Based on the above table, MU and SNDK show the highest growth potential with 81.5% and 84% gap in their Forward and Trailing P/E.

MU presents the strongest case, undervalued by both Method 1 and Method 2, with a huge Gap of 81.5% in Fwd and Trailing P/E Ratios, proving high market expectations of growth from the stock. With increased expectations emerging from short supply and continued demand for high-bandwidth memory (HBM) chips.

Despite being the most talked-about AI stock, NVDA is rated as undervalued under all 3 methods of valuation, suggesting that its profit growth has kept up with its price appreciation. Then comes SNDK, with the highest gap between Fwd and Trailing P/E of 84%, showcasing remarkable expectations of growth potential, however, it is important for readers to note that the lack of PEG data restricts the completeness of its assessment, and the conclusion undervalued is based on fewer data points than other companies.

PART 2: Asset Allocation and Portfolio Optimization

STEP 4: Previous price data and daily return analysis

Under this step, three years of daily closing prices were fetched for all the 5 companies using the yfinance library. Therefore, all the companies would have closing prices data of 750 trading days which will then form a base for further risk and volatility calculations.

Note:

SNDK recently went public and only has data available for 336 days. Therefore, to keep the comparison fair and objective all 5 companies are first assessed during the overlapping periods of 336 days only. SNDK's inclusion is justified by its remarkable fundamental profile, which includes the most rapid revenue growth of 251% and the strongest Forward vs. Trailing P/E gap of 84% among all screened businesses, even though a shorter data window diminishes statistical dependability.

To address this limitation, a separate analysis was conducted on the remaining four companies for 750 days, therefore producing a more statistically robust risk return metric.

Once the closing prices for all 336 days had been extracted, calculations were performed to present the daily percentage (%) change in closing prices of the stock for 335 days.

STEP 5: Derivation of 4 key Risk & Return measures from the daily percentage price change

Ticker	Annualized Return (%)	Annualized Volatility (%)	Sharpe Ratio	Beta
NVDA	41.51%	43.24%	0.86	1.83
MU	204.12%	71.34%	2.80	2.48
AVGO	48.65%	50.36%	0.88	1.88
LRCX	128.07%	55.09%	2.25	2.25
SNDK	352.44%	99.75%	3.49	2.68

Annualized Returns & Volatility

Annualized returns were calculated by simply deriving a mean of percentage price changes and multiplying it by 252 (number of trading days during a single year). This is an extrapolation, which assumes that if the pattern observed over the past 335 days were to continue consistently for the next full trading year, the result would give the annualized return figure. While this assumption will not hold in the practical world, it creates a good base for comparison of each stock's return potential.

Annualized Volatility was similarly computed by deriving the standard deviation of all the daily returns. The daily volatility was then scaled to an annual figure by multiplying it with the square root of 252, in accordance with square root of time rule.

Beta

Beta measures the sensitivity of each stock's returns relative to the broader market, represented by the S&P 500 index in the report. To calculate Beta, daily percentage price changes were first derived for S&P 500 over the same overlapping 335 trading day period for which our stock's have been evaluated. With both stock and market daily returns available, Covariance between each stock's returns and the S&P 500 returns was calculated, along with Variance of the S&P 500 returns alone. Beta was then derived using the standard formula,

Beta = Covariance (Stock, S&P 500) / Variance (S&P 500)

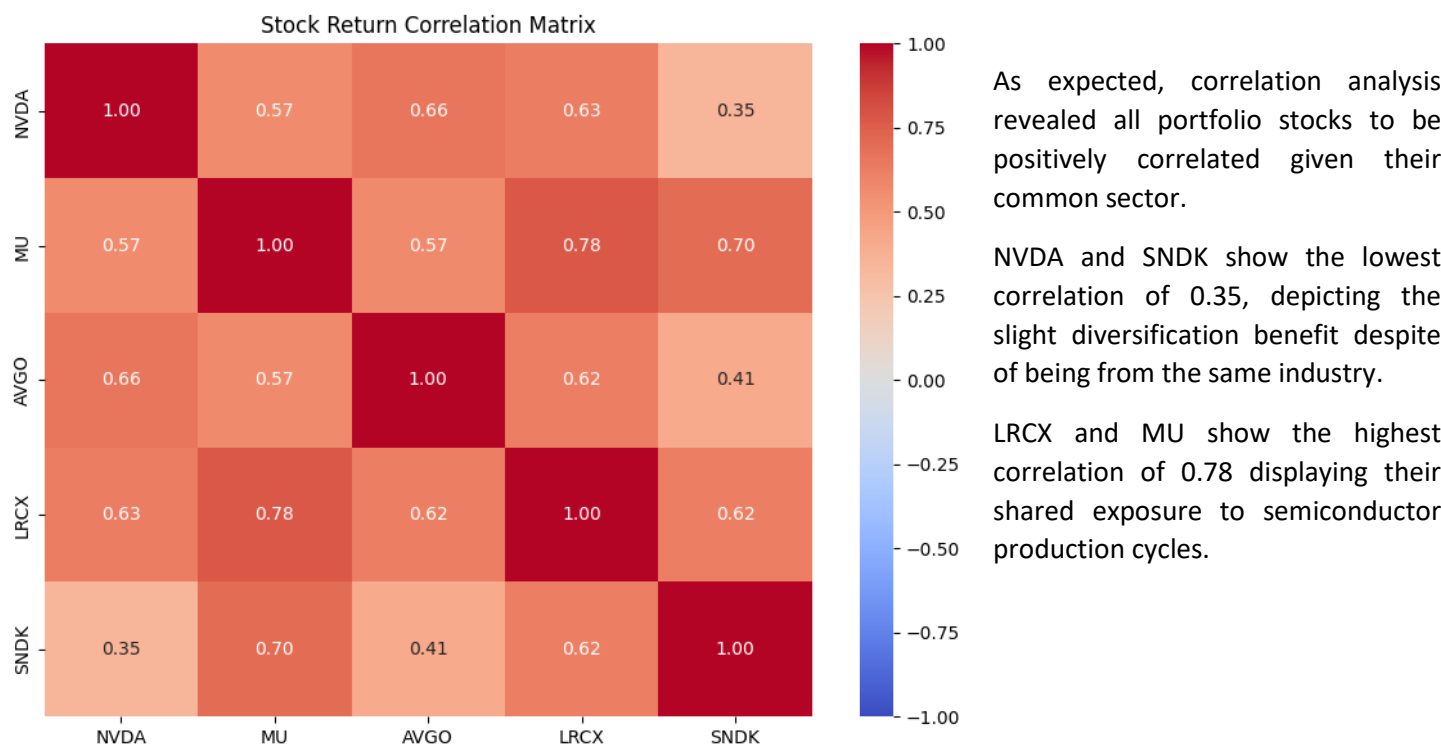
All five portfolio companies displayed high Beta values ranging from 1.83 to 2.68, indicating a substantially elevated sensitivity to market movements. This means that for every 1% move in the S&P 500, these stocks will move by 1.83% and 2.68% in the same direction. This portfolio is therefore most suitable for investors with high risk tolerance and a bullish market outlook. During downturns, this portfolio could result in losses double than that of the broader market.

Sharpe Ratio

The Sharpe Ratio is a measure of risk adjusted returns. The current US 10-year Treasury yield, or the return an investor might obtain with no risk, was used to determine the risk-free rate, which was set at 4.3%. The risk premium gained when opting to invest in stocks rather than risk-free government bonds is represented by any return above this cutoff.

Despite having a far lower public profile than NVDA and AVGO, risk-adjusted return analysis shows that SNDK and MU as the strongest performers, with Sharpe Ratios of 3.49 and 2.80, respectively. This conclusion is notably valuable as it proves that the most talked-about AI stocks are not always the best producers of risk-adjusted returns. Despite having excellent market reputations, NVDA and AVGO have Sharpe Ratios of 0.86 and 0.88, respectively, which are both below the 1.0 threshold that is regarded as strong in professional finance.

STEP 6: Computing Correlation



STEP 7: Running the Monte Carlo Simulation

The Monte Carlo simulation was used to determine the optimal asset allocation, which included the process of running 10,000 random combinations of allocations across all the five companies (normalized to sum to 100%). For each simulation, annualized return, volatility and Sharpe Ratio were calculated. The optimal portfolio was identified as the simulation with the highest Sharpe Ratio.

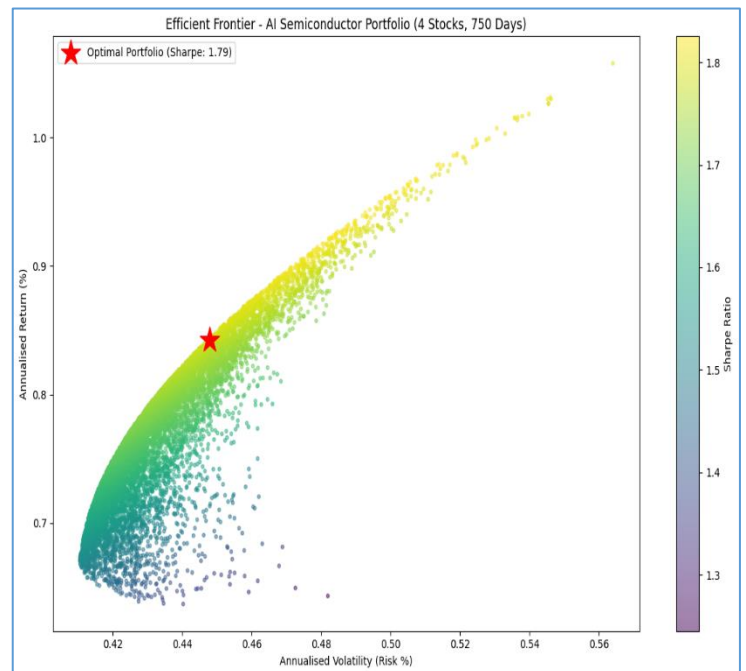
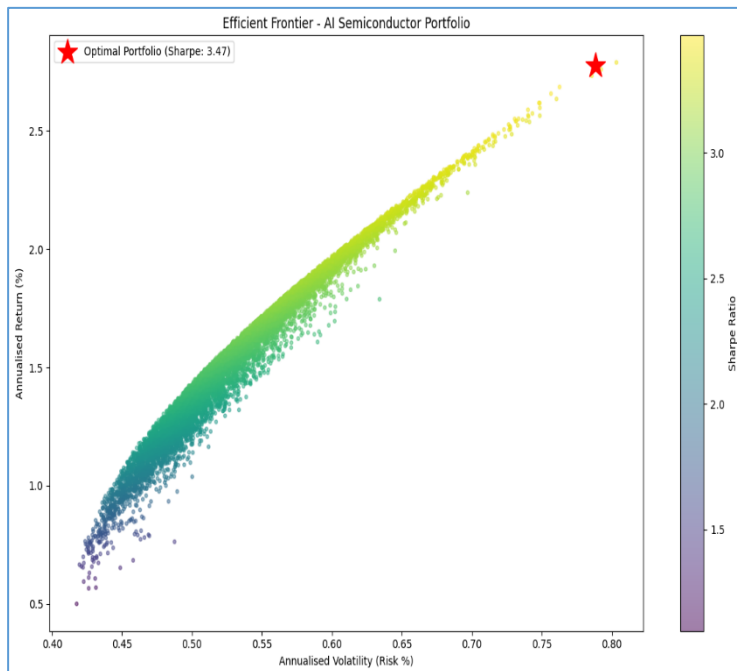
Two versions of the analysis were conducted:

- Version A: Included all 5 stocks (including SNDK), computed for 335 trading days
- Version B: Included only 4 stocks (excluding SNDK), computed for 750 trading days for greater statistical reliability

	VERSION A		VERSION B	
	5 Stocks Unconstrained	5 Stocks Constrained	4 Stocks Unconstrained	4 Stocks Constrained
NVDA	0.34%	5.00%	5.17%	27.83%
MU	36.28%	35.00%	55.64%	40.00%
AVGO	0.34%	5.00%	5.81%	8.63%
LRCX	8.38%	20.00%	33.38%	23.54%
SNDK	54.67%	35.00%	—	—
Expected Return	277.76%	224.92%	93.19%	83.94%
Volatility	78.48%	66.55%	48.23%	44.81%
Sharpe Ratio	3.48	3.31	1.89	1.78

Unconstrained vs Constrained

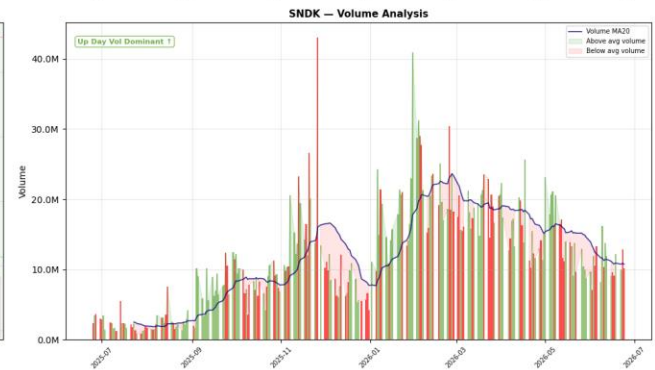
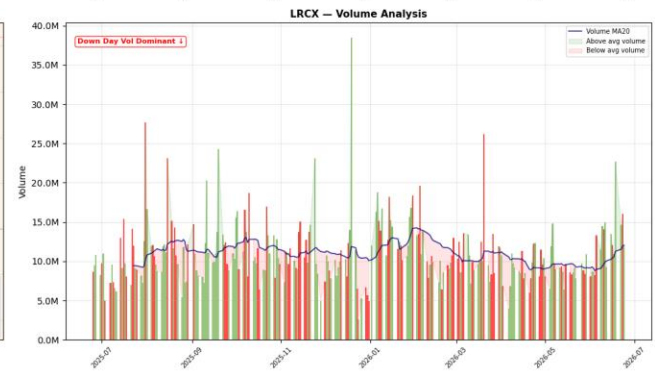
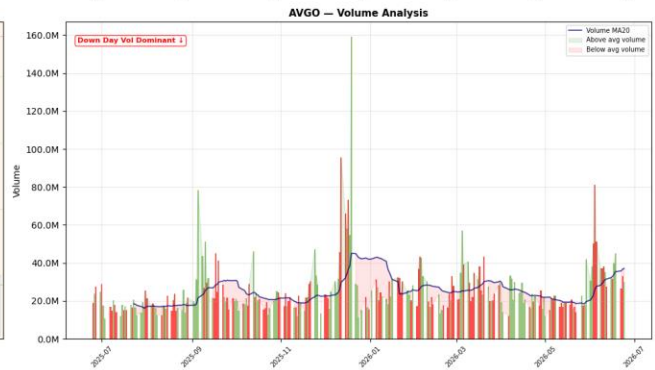
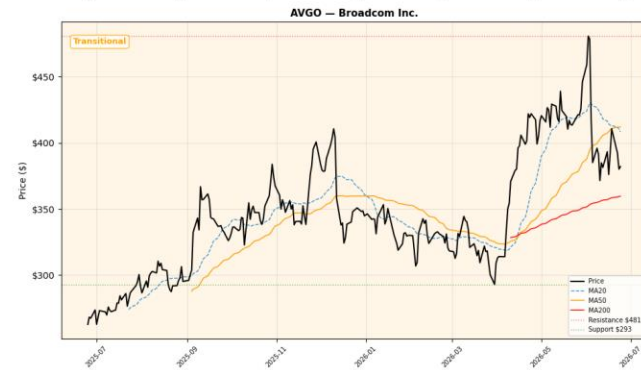
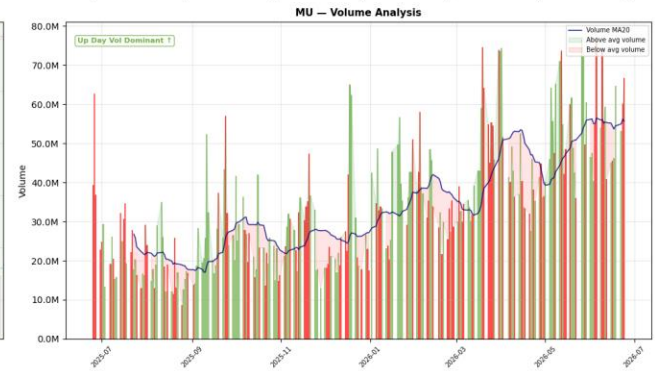
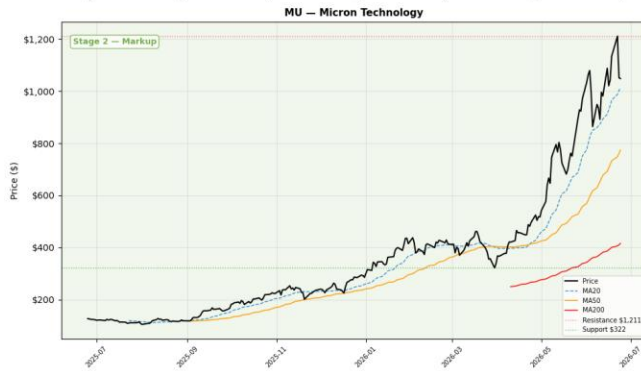
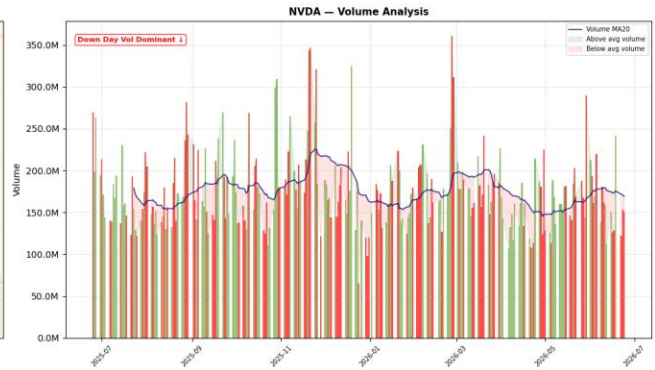
Unconstrained allocation allowed the code to allocate freely across all 5 company's stocks. While it found a combination with the highest sharpe ratio, the portfolio was highly concentrated with over 90% in just 2 companies (SNDK and MU) in the Version A of calculation. This would put the portfolio in a high-risk scenario, with almost no protection in times of adverse events. Therefore, constraints were introduced which places a minimum threshold of 5% and a maximum threshold of 35% per stock, for Version A only. This would lead to reduction in concentration of funds and diversify investment across all the companies. Similarly, a higher constraint of up to 40% was set in the Version B of calculations.



PART 3: Technical Analysis Tool

Technical Analysis Price Action & Volume 25 June 2026					
Metric	NVDA (NVIDIA Corporation)	MU (Micron Technology)	AVGO (Broadcom Inc.)	LRCX (Lam Research Corp.)	SNDK (Sandisk Corporation)
Current Price (\$)	\$199.00	\$1,048.51	\$382.07	\$374.80	\$1,914.46
MA20 (\$)	\$209.62	\$1,010.71	\$408.36	\$348.32	\$1,837.15
MA50 (\$)	\$210.05	\$773.66	\$412.02	\$305.29	\$1,469.77
MA200 (\$)	\$190.18	\$415.31	\$359.68	\$213.98	\$643.10
Price vs MA20	Below	Above	Below	Above	Above
Price vs MA50	Below	Above	Below	Above	Above
Price vs MA200	Above	Above	Above	Above	Above
Higher Highs	✓ Yes	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Higher Lows	X No	✓ Yes	X No	✓ Yes	✓ Yes
Volume Bias	Down Days	Up Days	Down Days	Down Days	Up Days
30D Return (%)	-9.76%	+36.78%	-8.73%	+29.67%	+31.85%
Prior 30D Return	+33.67%	+138.22%	+42.91%	+44.67%	+153.63%
Momentum	Slowing	Slowing	Slowing	Slowing	Slowing
MA20 Signal	Bearish ↓	Bullish ↑	Bearish ↓	Bullish ↑	Bullish ↑
MA50 Signal	Bearish ↓	Bullish ↑	Bearish ↓	Bullish ↑	Bullish ↑
MA200 Signal	Bullish ↑	Bullish ↑	Bullish ↑	Bullish ↑	Bullish ↑
Volume Signal	Bearish	Bullish	Bearish	Bearish	Bullish
Momentum Signal	Caution	Caution	Caution	Caution	Caution
WYCKOFF STAGE	Transitional	Stage 2 - Markup	Transitional	Late Stage 2 / Early-Stage 3	Stage 2 - Markup
OVERALL, ACTION	WAIT - Mixed Signals	BUY - Strong Uptrend	WAIT - Mixed Signals	HOLD - Monitor Closely	BUY - Strong Uptrend
Rationale	Could be in a pullback, wait for buying signals.	4/5 signals show a strong buy	Same as NVDA, wait for bounce back.	3/5 Bullish Signals	4/5 Signals present a strong buy

Price Action & Volume Analysis — AI Semiconductor Portfolio
Wyckoff Stage Classification | June 2026



PART 4: Forward Looking Analysis

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FORWARD LOOKING ANALYSIS – Thursday, 25 June 2026					
Source: Yahoo Finance via yfinance Data refreshed live					
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Metric	NVDA	MU	AVGO	LRCX	SNDK

— METHOD 1 – ANALYST PRICE TARGETS —					

Current Price (\$)	\$199.53	\$1,249.70	\$384.95	\$391.01	\$2,271.00
Target Low (\$)	\$180.00	\$249.00	\$215.88	\$220.00	\$1,000.00
Target Mean (\$)	\$298.93	\$1,048.21	\$523.84	\$340.58	\$1,751.32
Target High (\$)	\$500.00	\$1,750.00	\$650.00	\$480.00	\$3,250.00
Implied Return – Low (%)	-9.79%	-80.08%	-43.92%	-43.74%	-55.97%
Implied Return – Mean (%)	+49.82%	-16.12%	+36.08%	-12.90%	-22.88%
Implied Return – High (%)	+150.59%	+40.03%	+68.85%	+22.76%	+43.11%
Number of Analysts	59	39	45	31	22
Recommendation	STRONG_BUY	STRONG_BUY	STRONG_BUY	BUY	BUY
Analyst Signal	STRONG BUY	CAUTION	STRONG BUY	CAUTION	CAUTION

— METHOD 2 – FORWARD EARNINGS & REVENUE —					

Earnings Growth (%)	+214.50%	+756.00%	+85.40%	+40.80%	N/A
Revenue Growth (%)	+85.20%	+196.30%	+47.90%	+23.80%	+251.00%
Trailing EPS (\$)	\$6.53	\$44.25	\$6.00	\$5.28	\$29.29
Forward EPS (\$)	\$12.73	\$129.36	\$19.40	\$8.01	\$183.05
EPS Growth Implied (%)	+94.93%	+192.33%	+223.26%	+51.67%	+524.94%
Trailing P/E	30.56x	28.24x	64.16x	74.05x	77.48x
Forward P/E	15.68x	9.66x	19.85x	48.83x	12.40x
PEG Ratio	0.61x	0.31x	0.69x	2.18x	N/A
Fundamental Signal	STRONG BUY	STRONG BUY	STRONG BUY	STRONG BUY	BUY

— COMBINED CONCLUSION —					

COMBINED SIGNAL	STRONG BUY	BUY	STRONG BUY	BUY	NEUTRAL
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Signal Key: STRONG BUY = >20% upside + strong fundamentals					
BUY = >5% upside + decent fundamentals					
NEUTRAL = limited upside or mixed fundamentals					
CAUTION = negative targets or weak fundamentals					
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Conclusion & Recommendation

This report contains a 4 layered approach to evaluation of the AI sector, specifically diving into the semi-conductor Sub Sector.

PART 1&2 – Historical Analysis

5 major sub sectors in the AI industry were evaluated based on their ETF performances to find the best performing sub-sector, which concluded to be the Semi-Conductor Industry. Further fundamental screening was run across 34 companies in the SOXX to finalize the final 5 companies for further research, namely NVDA, AVGO, MU, SNDK and LRCX. Three separate methods of valuation assessment revealed that four of the five were undervalued, with MU and SNDK generating the strongest signals given their remarkable projected earnings growth of 756% and 525%, respectively. Monte Carlo optimisation of 10,000 simulations displayed MU and SNDK as strongest risk adjusted performers, with historical Sharpe Ratios of 2.80 and 3.49.

PART 3 – Technical Analysis

Wyckoff stage classification confirmed MU and SNDK to remain in stage 2 Markup, displaying intact uptrends with prices above MA50 and MA200, patterns showing higher highs and higher lows, and volume dominance on up days. NVDA and AVGO both showed transitional signals, both indicating short term pullbacks below MA50, but long-term uptrend above MA200.

PART 4 – Forward Looking Analysis

Using analyst's price targets and forward earnings & revenue, NVDA and AVGO were identified as strong buy opportunity, while MU and LRCX displayed buy decision. SNDK displaying a neutral signal, primarily due to insufficient PEG ratio data.

The Conflict

The most crucial output of this report is not any individual metric but it's the distinction between historical and forward-looking signals for MU and SNDK. While historical and technical analysis both conclude on the stocks being a strong buy, analyst's consensus price target implies a negative 12 month returns of -16% for MU and -22% for SNDK, indicating that the current prices may have already captured most of the near-term earnings growth. While this raises important considerations regarding entry timing at current price levels, it does not refute the long-term investing thesis.

Final Call

NVDA and AVGO are the only 2 stocks indicating consistent signals across all four analytical layers. Both exhibit positive forward earnings growth projections, carry strong analyst consensus targets of +49.82% and +36.08%, respectively, are fundamentally undervalued across three valuation methods, and are currently in technical pullbacks below MA50, which historically represents appealing entry opportunities in line with a long-term uptrend. NVDA and AVGO make the strongest and most consistent argument for investors wishing to act on this analysis at present, regardless of the method used.

Limitations

- Historical data: All return and risk metrics are calculated from historical price data. Past performance does not predict future results. The AI semiconductor sector experienced exceptional conditions during the analysis period that may not be repeated.
- Concentration risk: Despite being diversified across the semiconductor value chain, all 5 companies operate within a single sector. A sector-specific shock would impact all holdings simultaneously.
- No DCF analysis was conducted, future cash flows were not independently modelled.